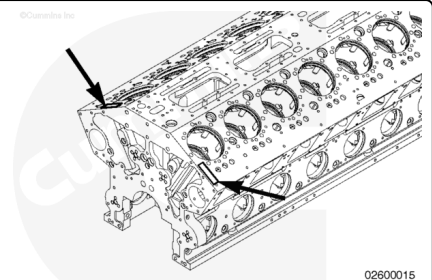


Trainings ▾

Install

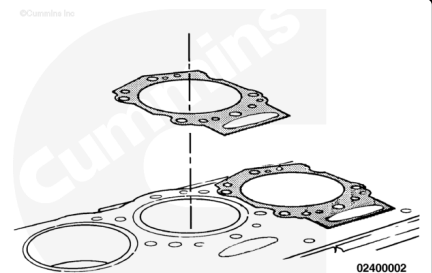
Inspect the cylinder block to determine if an oversize or a standard size head gasket is required.

An oversize head gasket is often required on blocks that have had material removed from the top surface of the block. Check the block for markings indicating an oversize gasket is required.



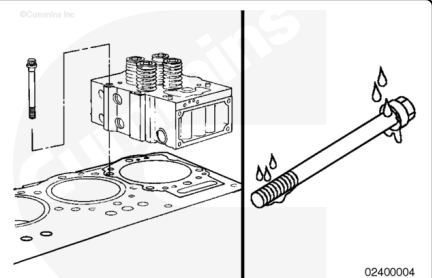
The word TOP, stamped on top of the head gasket, **must** be visible after the gasket is installed.

Install the cylinder head gasket, use the dowel pins for alignment.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the cylinder head. It **must** slide easily over the dowel pins.

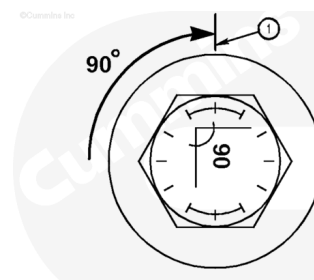
Lubricate the capscrew head flange with clean engine oil.

Lubricate the capscrew threads with clean engine oil. Allow excess oil to drip off the capscrews before installing them in the block.

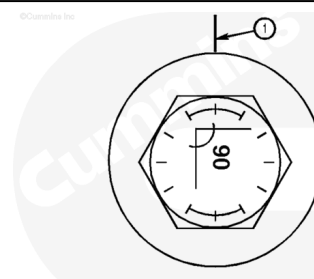
Install the capscrews.

When using torque plus angle, the tolerance on the 90-degree angle of rotation is 1½ flats (90 degrees ±5 degrees). If the capscrew is rotated beyond two flats, do **not** loosen the capscrew. The clamp load is still acceptable; however, rotating the capscrew beyond two flats causes additional stretch and reduces the life of the capscrew. With proper tightening, the capscrew can typically be reused for the life of the engine.

After all capscrews on one head have been tightened to 300 N•m [221 ft-lb], they **must** be rotated in the tightening direction an additional 90 degrees. Rotate the capscrew until the mark on the cylinder head (1) is between the next two marks joined by an arc (more than one flat and less than two flats).



The markings on the head of the flange head capscrews serve as an aid during installation. After tightening the capscrew to 300 N•m [221 ft-lb], mark the cylinder head (1) adjacent to one of the two single marks on the capscrew head.



Tighten the cylinder head mounting capscrews, use the following steps and in the sequence shown.

Torque Value:

1. 70 n•m [52 ft-lb]
2. 200 n•m [148 ft-lb]
3. 300 n•m [221 ft-lb]
4. Turn another 90 degrees (clockwise)

